

Åbne Kvantesystemer - Monte Carlo simulering

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1 Project Description

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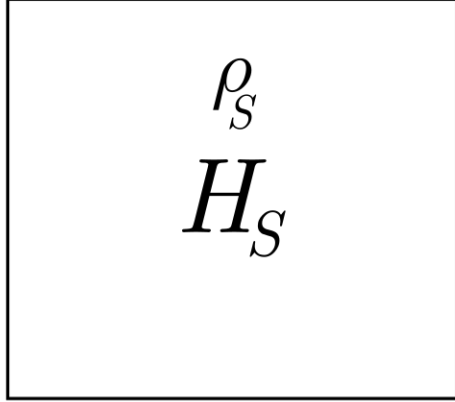
Projektbeskrivelse: Når vi realiserer kvantesystemer i den virkelige verden er de ikke helt alene. De vil altid være omringet af ting der kan interagere med dem, så som sortlegmestraling, andre partikler osv. Dette er blandt andet hvorfor en kvante-bit er svær at realisere uden at informationen i den går tabt. Det er umuligt for os at løse Schrödingerligningen for alt hvad der kan interagere med vores system eksakt, men ved at opskrive Schrödinger lignen med densitets formalisme kan man udlede Lindblad ligningen, som man ved hjælp af Monte Carlo simuleringer kan bruge til at beskrive ens system og miljø interaktion. I projektet vil i udlede densitets formalismens udgave af Schrödinger ligningen (Von Neumann ligningen), i vil fortolke Lindblad ligningen, samt Monte Carlo simulere et to niveau system (en kvante-bit) der interagerer med sortlegmestraling.

2 Introduction - Density Matrix Formalism

In this project we will try to describe a system which interacts with an environment. So far in Griffiths we have always assumed that all of our systems, be they the Harmonic oscillator, the Hydrogen atom or any other, that they are the only thing that exist. Therefore, we have always solved the Hamiltonian $H = H_{System}$. Solving for the system Hamiltonian (and in general for a Hamiltonian which we can write up in terms of operators) is what we call a *closed quantum system*. However, in reality all systems exist in an environment and in principle we should always consider $H = H_{System} + H_{Environment} + H_{Interaction}$. The difficult part is that we typically don't know how to write up the Hamiltonian for the environment, which makes our system an *open quantum system* (see figure 2). In this project we will deal with open quantum systems. This note should be self containing, but you might want to use additional reading. The book Modern Atomic and Optical Physics II chapters 3 and 6 are nice, but a bit advanced. This first section will primarily work with the theoretical description of an open quantum system, while the next section will describe the method of stochastic wavefunctions with numerical applications, which is the primary goal of the project. Thus, read this section lightly and focus on the key result (the Lindblad equation), which we need for the stochastic Monte Carlo method.

For a closed quantum system Hamiltonian H_S , you have seen in this course that we can solve the Schrödinger equation and find the wavefunctions and energies that describe the system which is described by H_S . However, to proceed with our description of open quantum systems, we have to abandon some information of our system. Instead of knowing every wavefunction exactly, we turn

Closed Quantum System



Open Quantum System

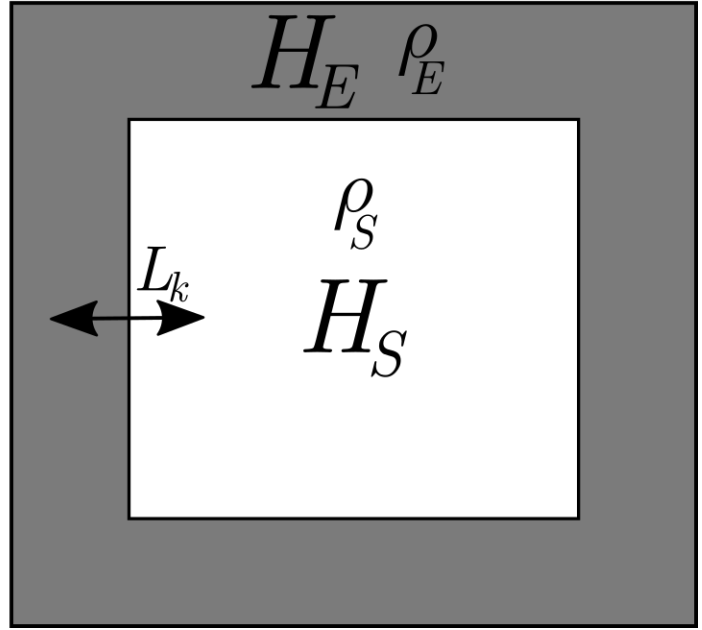


Figure 1: Left: A depiction of a closed quantum system completely described by a known system Hamiltonian H_S . Right: A depiction of an open quantum system described by a known system Hamiltonian H_S and an unknown environment Hamiltonian H_E using the Lindblad equation and the exchange operators L_k

to a picture of a density of wavefunctions,

$$\rho = |\Psi\rangle\langle\Psi| \quad (1)$$

The density ρ described like this is an operator in quantum mechanics, but it does behave like a classical density, when you consider the matrix elements of it.

- Show that for two stationary solutions ($|\psi_1\rangle$, $|\psi_2\rangle$) to a system Hamiltonian H_S the wavefunction,

$$|\Psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle ,$$

has density matrix elements

$$\begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} = \begin{pmatrix} |c_1|^2 & c_1 c_2^* \\ c_1^* c_2 & |c_2|^2 \end{pmatrix}$$

. Remember $\rho_{ij} = \langle\psi_i|\rho|\psi_j\rangle$.

Thus the diagonal of the density matrix describe the probability (density) of being in a specific stationary state (normal mode) and the off-diagonal elements describe the phase relation between those states, which is analogous to a density of states in the normal mode description of classical mechanics.

Let us for now only consider the density described by the system, which we denote ρ_S . To abandon the wavefunction picture in favour of a density picture, we thus wish to write up the Schrödinger

equation with our density operator,

$$i\hbar \frac{\partial}{\partial t} \rho_S = i\hbar \frac{\partial}{\partial t} |\Psi\rangle\langle\Psi| = i\hbar \left| \frac{\partial}{\partial t} \Psi \right\rangle \langle\Psi| + i\hbar |\Psi\rangle \left\langle \frac{\partial}{\partial t} \Psi \right| \quad (2)$$

$$i\hbar \left| \frac{\partial}{\partial t} \Psi \right\rangle = H_S |\Psi\rangle \quad (3)$$

$$\overline{i\hbar \left| \frac{\partial}{\partial t} \Psi \right\rangle} = -i\hbar \left\langle \frac{\partial}{\partial t} \Psi \right| = \langle\Psi| H_S \quad (4)$$

$$(5)$$

- Combine the above equations to derive the Von Neumann equation,

$$\frac{\partial}{\partial t} \rho_S = -\frac{i}{\hbar} [H_S, \rho]. \quad (6)$$

The Von Neumann equations is what is known as the quantized Liouville equation, which in classical mechanics read,

$$\frac{\partial}{\partial t} \rho_S = \{H_S, \rho\}. \quad (7)$$

The Liouville equation describes the trajectory of a density of states in classical mechanics and we have now derived a quantum mechanical version of this equation. If we choose an arbitrary basis $\{|\psi_i\rangle, i = 1 \dots N\}$ to describe our density matrix elements in we can solve the Von Neumann equation without ever knowing the correct basis of stationary states to H_S . Thus, we have sacrificed knowing the information of the wavefunction solutions to H_S , for knowing the solution to the density ρ_S . This is a bit silly if we have a system Hamiltonian H_S for which we can solve, but this turns out to be a strong tool when we expand to a Hamiltonian including the environment, which we can not describe or solve.

- For the two level system previously considered

$$|\Psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle,$$

let the Hamiltonian be the Harmonic oscillator $H_S = \hbar\omega (a_+ a_- + \frac{1}{2})$. Find the time evolution of the density operator $\partial/\partial t \rho_S$. Answer given as the matrix elements of $\dot{\rho}$:

$$\begin{pmatrix} \dot{\rho}_{11} & \dot{\rho}_{12} \\ \dot{\rho}_{21} & \dot{\rho}_{22} \end{pmatrix} = -i\omega \begin{pmatrix} 0 & -c_1 c_2^* \\ c_1^* c_2 & 0 \end{pmatrix}.$$

The probability of being in a specific state never changes $\dot{\rho}_{11} = \dot{\rho}_{22} = 0$, as we would expect. If we prepare a state $|\Psi\rangle$ as a linear combination of stationary solutions, the state is a stationary solution. But the phase between the two states (the wiggle factor) rotates with time, as we expect from the solutions we have using the Schrödinger equation.

- (Optional) Let us make it more interesting and introduce two operators

$$\sigma_+ = |\psi_2\rangle\langle\psi_1|, \quad \sigma_- = |\psi_1\rangle\langle\psi_2|.$$

These operators either raise (σ_+) $|\psi_1\rangle$ to state $|\psi_2\rangle$ or lower (σ_-) state $|\psi_2\rangle$ to state $|\psi_1\rangle$. Let us then adjust our harmonic oscillator Hamiltonian to be,

$$H = \hbar\omega \left(a_+ a_- + \frac{1}{2} \right) + \hbar\lambda (\sigma_+ + \sigma_-).$$

It can now be showed that

$$\begin{pmatrix} \dot{\rho}_{11} & \dot{\rho}_{12} \\ \dot{\rho}_{21} & \dot{\rho}_{22} \end{pmatrix} = -i\omega \begin{pmatrix} 0 & -c_1 c_2^* \\ c_1^* c_2 & 0 \end{pmatrix} - i\lambda \begin{pmatrix} c_1^* c_2 - c_1 c_2^* & |c_2|^2 - |c_1|^2 \\ |c_1|^2 - |c_2|^2 & c_1 c_2^* - c_1^* c_2 \end{pmatrix}.$$

Show that the above equation is true. For an initial condition $c_2(0) = 1$ and $c_1(0) = 0$, the solution to the above is (you don't have to solve this),

$$c_2(t) = \cos(\lambda t) e^{-i\frac{\omega}{2}t}, \quad (8)$$

$$c_1(t) = -i \sin(\lambda t) e^{i\frac{\omega}{2}t}. \quad (9)$$

This result is an example of the Jaynes-Cummings model, which describes the interaction between a two-level system and a light field, which excites population from the excited state to the ground state and back again. Figure 2 shows the behavior of the probability of being in either state $|\psi_1\rangle$ or $|\psi_2\rangle$ as a function of time. If we tune λt properly we can move the population from being completely in a state $|\psi_2\rangle$ to $|\psi_1\rangle$ and vice versa. Since these states represent information in a "binary" 0-1 state we have created a quantum bit, which is what is used in quantum computers.

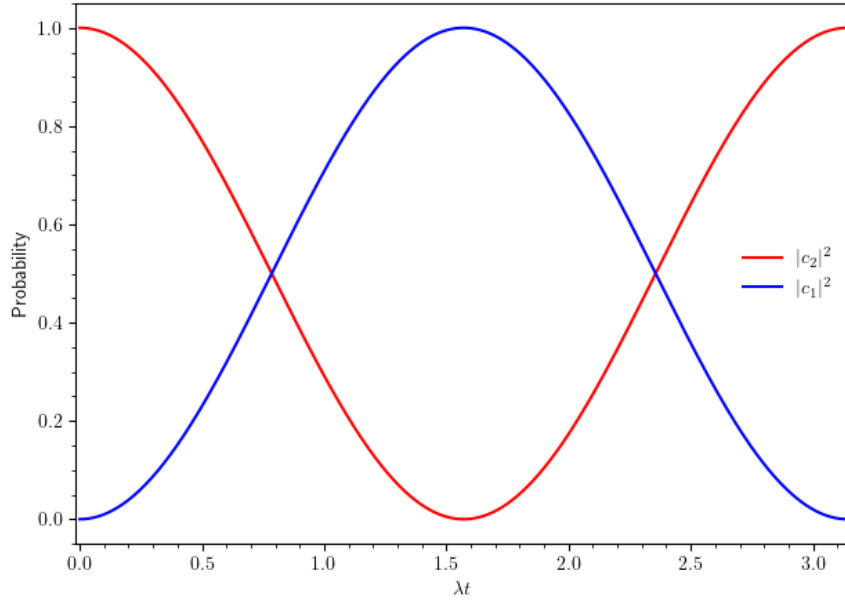


Figure 2: Jaynes-Cummings two-level model with a classical light field.

2.1 The Lindblad Equation

Now that we have build some understanding of the Von Neumann equation we turn on the environment and the interaction between the environment and the system. However to proceed with our nem Hamiltonian $H = H_S + H_E + H_I$ we need a couple of tricks and assumptions. This is a lot of math and if it does not interest you, gloss lightly over the derivation and skip to equation 37.

2.1.1 Derivation of the Lindblad Equation (the Quantum Master Equation)

Let us assume that when we begin our time keeping $t = 0$ the density of the system and the environment decouple (they are not correlated),

$$\rho(t=0) = \rho_S(t=0)\rho_E(t=0). \quad (10)$$

Let us introduce a new type of operation we can do on a quantum operator. If we know a basis of wavefunctions which span a subset of our Hamiltonian we can *trace* out those degrees of freedom. For example, assume that we know the basis wavefunctions which describe the system $\{\psi_i\}$ and the basis wavefunctions which describe the environment $\{\chi_i\}$ we can write the density as

$$\rho_S(t) = \sum_{i \in \text{System}} |\psi_i\rangle\langle\psi_i|, \quad (11)$$

$$\rho_E(t) = \sum_{i \in \text{Environment}} |\chi_i\rangle\langle\chi_i|. \quad (12)$$

We now perform the trace of $\rho(t)$ with respect to the environment as,

$$\text{tr}_E \rho(t) = \sum_{i \in \text{Environment}} \langle\chi_i(t)|\rho(t)|\chi_i(t)\rangle = \rho_S(t). \quad (13)$$

This has removed the environment degrees of freedom. It is in general true that $\text{tr}_E \rho(t) = \rho_S(t)$, but in the second part of the above, we have made the Born approximation, which states that the Environment act as an infinitely large bath, such that the interaction of the system with the environment never alters the state of the environment. Thus it is not only true that $\rho(t)$ is factorable for $t = 0$, but for all times $\rho(t) = \rho_S(t)\rho_E$ (sometimes you will encounter ρ_E written as R_0 to symbolize that it is time independent).

- Prove equation 13 under the Born approximation

Now we introduce a useful trick. Transform all operators into an interaction picture by defining for an operator X ,

$$U_0 = e^{\frac{i}{\hbar}(H_S + H_E)t}, \quad \tilde{X} = U_0 X U_0^\dagger. \quad (14)$$

We then write the Von Neumann equation for our density $\rho(t)$ as

$$\dot{\rho} = \dot{U}_0 \rho U_0^\dagger + U_0 \dot{\rho} U_0^\dagger + U_0 \rho \dot{U}_0^\dagger, \quad (15)$$

$$= \frac{i}{\hbar} (H_S + H_E) U_0 \rho U_0^\dagger - \frac{i}{\hbar} U_0 \rho (H_S + H_E) U_0^\dagger - \frac{i}{\hbar} U_0 [H, \rho] U_0^\dagger, \quad (16)$$

$$= \frac{i}{\hbar} U_0 [H_S + H_E, \rho] U_0^\dagger - \frac{i}{\hbar} U_0 [H, \rho] U_0^\dagger, \quad (17)$$

$$= -\frac{i}{\hbar} U_0 [H_I, \rho] U_0^\dagger, \quad (18)$$

$$= -\frac{i}{\hbar} [\tilde{H}_I, \tilde{\rho}]. \quad (19)$$

This is then the Von Neumann in the interaction picture. Before we just introduce our density $\rho(t) = \rho_S(t)\rho_E$ we first massage the corresponding $\tilde{\rho}(t)$,

$$\tilde{\rho}(t) = \tilde{\rho}(0) + \int_0^t dt' \dot{\tilde{\rho}}(t'), \quad (20)$$

$$= \tilde{\rho}(0) - \frac{i}{\hbar} \int_0^t dt' [\tilde{H}_I(t'), \tilde{\rho}(t')]. \quad (21)$$

We used the Von Neumann equation in the interaction picture to derive the above, but now we put the above back into the Von Neumann equation in the interaction picture,

$$\dot{\tilde{\rho}}(t) = -\frac{i}{\hbar} [\tilde{H}_I(t), \tilde{\rho}(0)] - \frac{1}{\hbar^2} \int_0^t dt' [\tilde{H}_I(t), [\tilde{H}_I(t'), \tilde{\rho}(t')]] . \quad (22)$$

$$\text{tr}_E \{\dot{\tilde{\rho}}(t)\} = \text{tr}_E \left\{ \frac{i}{\hbar} [\tilde{H}_I(t), \tilde{\rho}(0)] \right\} - \frac{1}{\hbar^2} \int_0^t dt' \text{tr}_E \left\{ [\tilde{H}_I(t), [\tilde{H}_I(t'), \tilde{\rho}(t')]] \right\} , \quad (23)$$

$$\dot{\rho}_S(t) = -\frac{1}{\hbar^2} \int_0^t dt' \text{tr}_E \left\{ [\tilde{H}_I(t), [\tilde{H}_I(t'), \tilde{\rho}_S(t') \rho_E]] \right\} . \quad (24)$$

Were we used the Born approximation from the second to the third line, which states that H_I doesn't affect the environment (which removes the first term) and that the density therefore decouples to $\rho(t) = \rho_S(t) \rho_E$. We make one last assumption that the time derivative of the system density only depends on its current state and that all correlation between the current state and previous states are lost. Using this approximation (Born-Markov approximation) we can change $\tilde{\rho}_S(t') \rightarrow \tilde{\rho}_S(t)$ in the above. In principle we have now arrived at what is called the master equation of quantum mechanics, but we want a prettier equation to work with, so we assume that the interaction hamiltonian can be written as,

$$H_I = \sum_i \alpha_i(t) \Gamma_i^\dagger(t) + \Gamma_i(t) \alpha_i^\dagger(t), \quad (25)$$

$$\tilde{H}_I = \sum_i \tilde{\alpha}_i(t) \tilde{\Gamma}_i^\dagger(t) + \tilde{\Gamma}_i(t) \tilde{\alpha}_i^\dagger(t), \quad (26)$$

where α_i is an operator of the system and Γ_i an operator of the environment. You can interpret it as $\Gamma_i^\dagger$ takes out energy of the environment and α_i then puts that into the system and vice versa for $\Gamma_i \alpha_i^\dagger$.

$$\dot{\rho}_S(t) = -\frac{1}{\hbar^2} \sum_{i,j} \int_0^t dt' (\tilde{\alpha}_i(t) \tilde{\alpha}_j^\dagger(t') \tilde{\rho}_S(t') - \tilde{\alpha}_i(t) \tilde{\rho}_S(t') \tilde{\alpha}_j^\dagger(t')) \text{tr}_E \left\{ \Gamma_i^\dagger(t) \Gamma_j^\dagger(t') \rho_E \right\} + \text{h.c.} \quad (27)$$

$$+ (\tilde{\alpha}_i^\dagger(t) \tilde{\alpha}_j^\dagger(t') \tilde{\rho}_S(t') - \tilde{\alpha}_i^\dagger(t) \tilde{\rho}_S(t') \tilde{\alpha}_j^\dagger(t')) \text{tr}_E \left\{ \Gamma_i(t) \Gamma_j(t') \rho_E \right\} + \text{h.c.} \quad (28)$$

$$+ (\tilde{\alpha}_i^\dagger(t) \tilde{\alpha}_j(t') \tilde{\rho}_S(t') - \tilde{\alpha}_i(t) \tilde{\rho}_S(t') \tilde{\alpha}_j^\dagger(t')) \text{tr}_E \left\{ \Gamma_i^\dagger(t) \Gamma_j(t') \rho_E \right\} + \text{h.c.} \quad (29)$$

$$+ (\tilde{\alpha}_i(t) \tilde{\alpha}_j^\dagger(t') \tilde{\rho}_S(t') - \tilde{\alpha}_i^\dagger(t) \tilde{\rho}_S(t') \tilde{\alpha}_j(t')) \text{tr}_E \left\{ \Gamma_i(t) \Gamma_j^\dagger(t') \rho_E \right\} + \text{h.c.} \quad (30)$$

The terms including $\Gamma_i^\dagger \Gamma_j^\dagger$ (or the conjugate thereof) are second order terms in the interaction and are disregarded. $\Gamma_i \Gamma_j^\dagger$ is very reminiscent of the counting operator in a Harmonic oscillator ($a_+ a_-$), so let us assume that the interaction instantaneously losses all correlation (Wigner-Weisskopf approximation) and that the strength of the interaction is $\gamma_{i,j}$. Then,

$$\text{tr}_E \left\{ \Gamma_i(t) \Gamma_j^\dagger(t') \rho_E \right\} = \hbar^2 \delta(t - t') \frac{\gamma_{i,j}}{2} \bar{n} \quad (31)$$

$$\text{tr}_E \left\{ \Gamma_i^\dagger(t) \Gamma_j(t') \rho_E \right\} = \hbar^2 \delta(t - t') \frac{\gamma_{i,j}}{2} (\bar{n} + 1) \quad (32)$$

This was a lot of math, but we finally arrive at the equation,

$$\dot{\rho}_S(t) = - \sum_{i,j} \frac{\gamma_{i,j}}{2} (\bar{n} + 1) (\tilde{\alpha}_i^\dagger \tilde{\alpha}_j \tilde{\rho}_S - \tilde{\alpha}_i \tilde{\rho}_S \tilde{\alpha}_j^\dagger) + \text{h.c.}, \quad (33)$$

$$- \sum_{i,j} \frac{\gamma_{i,j}}{2} \bar{n} (\tilde{\alpha}_i \tilde{\alpha}_j^\dagger \tilde{\rho}_S - \tilde{\alpha}_i^\dagger \tilde{\rho}_S \tilde{\alpha}_j) + \text{h.c.} \quad (34)$$

$$\simeq \sum_{i=j} \gamma_i \bar{n} \tilde{\alpha}_i \tilde{\rho}_S \tilde{\alpha}_i^\dagger - \frac{\gamma_i}{2} \bar{n} \{ \tilde{\alpha}_i^\dagger \tilde{\alpha}_i, \tilde{\rho}_S \} + \text{h.c.} \quad (35)$$

Where we at the last line assume that only one interaction at a time takes place $\gamma_{i,j} = 0$ for $i \neq j$ and that $\bar{n}$ is large. We now collect the terms $\gamma_i \bar{n} \alpha_i$ into the lindblad operator L_i ($\gamma_i \bar{n} \alpha_i = L_i$), such that returning to the Schödinger picture we have the Lindblad equation,

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar} [H_S, \rho_S] + \sum_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho_S \} \right) + \text{h.c.} \quad (36)$$

In fact, we did not have to assume $\bar{n} + 1 \simeq 1$ and we can negelect the addition of the h.c. if we make a proper definition of L_k and the most general form of the Lindblad equation is given in the beginning of the next section (see chapter 6 of Moden Atomic and Optical Physics II).

2.1.2 Interpretation of the Lindblad Equation

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar} [H_S, \rho_S] + \sum_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho_S \} \right) \quad (37)$$

With a lot of approximations and tears we have finally found the master equation (the Lindblad equation) which we have been searching for. Let us recount what we have done: we have assumed that the environment act as an infinitely large bath, which never changes and destroys any information (correlation) of our system at all previous times. The Lindblad operators L_k are system operators, which act as the operators that "exchange" energy between the environment and the system (see figure 2). What these operators are, depends on the interaction between the system and the environment, and we will deal with some examples of these in this project.

- (Optional) In the two-level Harmonic oscillator system we previously considered, we introduced a system Hamiltonian,

$$H_S = \hbar\omega \left(a_+ a_- + \frac{1}{2} \right) + \hbar\lambda(\sigma_+ + \sigma_-).$$

The last part of this Hamiltonian was us interacting with the system by the use of light interaction. We saw that by tuning λt we could move population from one state into the other (see figure 2). Now we wish to further develop this, by allowing our two-level system to interact with a thermal bath (an environment which decohere our system). Let us define the Lindblad operators as

$$L_{2k} = \gamma_k \sqrt{\bar{n} + 1} \sigma_-, \\ L_{2k+1} = \gamma_k \sqrt{\bar{n}} \sigma_+.$$

That is, we have defined an environment which either takes or gives energy to our system. We previously dealt with the Von Neumann term of the Lindblad equation and now complete our description of the open two-level system with $k = 0$,

$$\begin{aligned} \begin{pmatrix} \dot{\rho}_{11} & \dot{\rho}_{12} \\ \dot{\rho}_{21} & \dot{\rho}_{22} \end{pmatrix} = & -i\omega \begin{pmatrix} 0 & -c_1 c_2^* \\ c_1^* c_2 & 0 \end{pmatrix} \\ & -i\lambda \begin{pmatrix} c_1^* c_2 - c_1 c_2^* & |c_2|^2 - |c_1|^2 \\ |c_1|^2 - |c_2|^2 & c_1 c_2^* - c_1^* c_2 \end{pmatrix} \\ & -\gamma \begin{pmatrix} \bar{n}|c_1|^2 - (\bar{n}+1)|c_2|^2 & (\bar{n} + \frac{1}{2}) c_1 c_2^* \\ (\bar{n} + \frac{1}{2}) c_1^* c_2 & (\bar{n}+1)|c_2|^2 - \bar{n}|c_1|^2 \end{pmatrix}. \end{aligned} \quad (38)$$

These are the optical Bloch equations. Show this result by insertion in the Lindblad equation 37. The interaction with the environment act as excitations $\gamma\bar{n}$ and spontaneous emission γ of the system. You will encounter these equations in other courses, but here we recount that if we use a strong electric field (an intense photon beam), we get that $\rho_{22} \rightarrow \frac{1}{2}$ as time progresses, but for a weak electric field the environment dominates and $\rho_{22} \rightarrow 0$. Thus no matter how well we shield a two-level system from an environment (the electric field strength much bigger than the environment strength) we can at best achieve $\rho_{22}(t \rightarrow \infty) = \frac{1}{2}$. When $\rho_{22} = \frac{1}{2}$ we have lost the information we initially stored in the system $\rho_{22}(0) = 1$. The timescale this loss of information happens on is called the decoherence timescale of the system-environment. It is the golden variable to optimize in quantum computing.

3 Stochastic Wavefunctions - Monte Carlo Simulations

Let us write up the density operator of our system in some basis $\{|\psi_i\rangle\}$ and weights $\{p_i\}$ (probability density of a given wavefunction $|\psi_i\rangle$) as,

$$\rho_S = \sum_i p_i |\psi_i\rangle\langle\psi_i|. \quad (39)$$

Now we rewrite the Lindblad equation 37 to a different form,

$$\frac{d\rho_S}{dt} = \sum_i -\frac{i}{\hbar} \left[H_S - \frac{i\hbar}{2} \sum_k L_k^\dagger L_k \right] p_i |\psi_i\rangle\langle\psi_i| \quad (40)$$

$$+ \frac{i}{\hbar} p_i |\psi_i\rangle\langle\psi_i| \left[H_S + \frac{i\hbar}{2} \sum_k L_k^\dagger L_k \right] \quad (41)$$

$$+ \sum_k p_i L_k |\psi_i\rangle\langle\psi_i| L_k^\dagger, \quad (42)$$

$$\frac{d\rho_S}{dt} = \sum_i p_i \left[\frac{-i}{\hbar} \left(H_{eff} |\psi_i\rangle\langle\psi_i| - |\psi_i\rangle\langle\psi_i| H_{eff}^\dagger \right) + \sum_k L_k |\psi_i\rangle\langle\psi_i| L_k^\dagger \right]. \quad (43)$$

Here we defined an effective Hamiltonian

$$H_{eff} = H_S - \frac{i\hbar}{2} \sum_k L_k^\dagger L_k.$$

The first two terms of this "new" master equation resembles that of a pure state evolving according to a new Hamiltonian $H_S \rightarrow H_{eff}$. However, the last term we will interpret as a *quantum jump* operator which changes a state $|\psi_i\rangle \rightarrow |\phi_{k,i}\rangle = L_k |\psi_i\rangle$, with some probability. So with some given probability we either evolve a state according to H_{eff} or we make the state jump into another state $|\phi_{k,i}\rangle$. Our job now is to describe this stochastically (in small discrete time steps). We will therefore focus on the two parts separately first. The Schrödinger equation for a state evolving via H_{eff} reads,

$$i\hbar \frac{\partial}{\partial t} |\psi_i\rangle = H_{eff} |\psi_i\rangle. \quad (44)$$

Numerically we want to introduce a small timestep Δt and evolve a state by computing (this is the stochastic process),

$$|\psi_i(t + \Delta t)\rangle = \left(1 + \frac{H_{eff}\Delta t}{i\hbar}\right) |\psi_i(t)\rangle. \quad (45)$$

However, this wavefunction is not normalized, since the effective Hamiltonian is not hermitian. Therefore, we evaluate $\langle\psi_i(t + \Delta t)|\psi_i(t + \Delta t)\rangle$,

$$\langle\psi_i(t + \Delta t)|\psi_i(t + \Delta t)\rangle = \langle\psi_i|\left(1 - \frac{H_{eff}^\dagger\Delta t}{i\hbar}\right)\left(1 + \frac{H_{eff}\Delta t}{i\hbar}\right)|\psi_i\rangle \quad (46)$$

$$= \langle\psi_i|1 - \frac{\Delta t}{i\hbar}(H_{eff}^\dagger - H_{eff})|\psi_i\rangle \quad (47)$$

$$= \langle\psi_i|1 - \Delta t \sum_k L_k^\dagger L_k|\psi_i\rangle. \quad (48)$$

To simplify this we introduce the notation,

$$\Delta p_{k,i} = \Delta t \langle\psi_i|L_k^\dagger L_k|\psi_i\rangle, \quad (49)$$

$$\Delta p_i = \sum_k \Delta p_{k,i}. \quad (50)$$

With this notation we can write,

$$\langle\psi_i(t + \Delta t)|\psi_i(t + \Delta t)\rangle = 1 - \Delta p_i. \quad (51)$$

The normalization of the wavefunction once we have let it evolve via H_{eff} is this,

$$|\tilde{\psi}_i(t + \Delta t)\rangle = \frac{1}{\sqrt{1 - \Delta p_i}} \left(1 + \frac{H_{eff}\Delta t}{i\hbar}\right) |\psi_i(t)\rangle. \quad (52)$$

If we wanted to now, we could make a numerical program where we for each timestep Δt evolved any basis function according to H_{eff} and then normalized it with $\sqrt{1 - \Delta p_i}$. But we still need to include our quantum jump states $|\phi_{k,i}\rangle$. We normalize those states equivalently by,

$$\langle\phi_{k,i}|\phi_{k,i}\rangle = \langle\psi_i|L_k^\dagger L_k|\psi_i\rangle, \quad (53)$$

$$|\tilde{\phi}_{k,i}(t + \Delta t)\rangle = \sqrt{\frac{\Delta t}{\Delta p_{k,i}}} |\phi_{k,i}(t)\rangle. \quad (54)$$

Now the only thing left for us to do is to combine both the evolution of the states $|\psi_i\rangle$ and the quantum jump states $|\phi_{k,i}\rangle$ in the Lindblad equation

$$\begin{aligned} \rho_S(t + \Delta t) = \sum_i p_i \left[(1 - \Delta p_i) \left| \tilde{\psi}_i(t + \Delta t) \right\rangle \left\langle \tilde{\psi}_i(t + \Delta t) \right| \right. \\ \left. + \sum_k \Delta p_{k,i} \left| \tilde{\phi}_{k,i}(t + \Delta t) \right\rangle \left\langle \tilde{\phi}_{k,i}(t + \Delta t) \right| \right]. \end{aligned} \quad (55)$$

We have obtained a probabilistic master equation, which now tells us what the probability of a specific outcome after Δt time has passed is.

- With a probability $(1 - \Delta p_i)$ the system evolves according to H_{eff} and remains in state $|\tilde{\psi}_i\rangle$
- With probability $\Delta p_{k,i}$ the system jumps into another state $|\tilde{\phi}_{k,i}\rangle$. The total probability of jumping is Δp_i
- (Task) prove that the total sum of probabilities is 1

With this probabilistic stochastic picture of the evolution of a system wavefunction which interacts with an environment, we are now able to provide an algorithm for how to evolve a specific state $|\psi_i\rangle$. This is done with a Monte Carlo procedure,

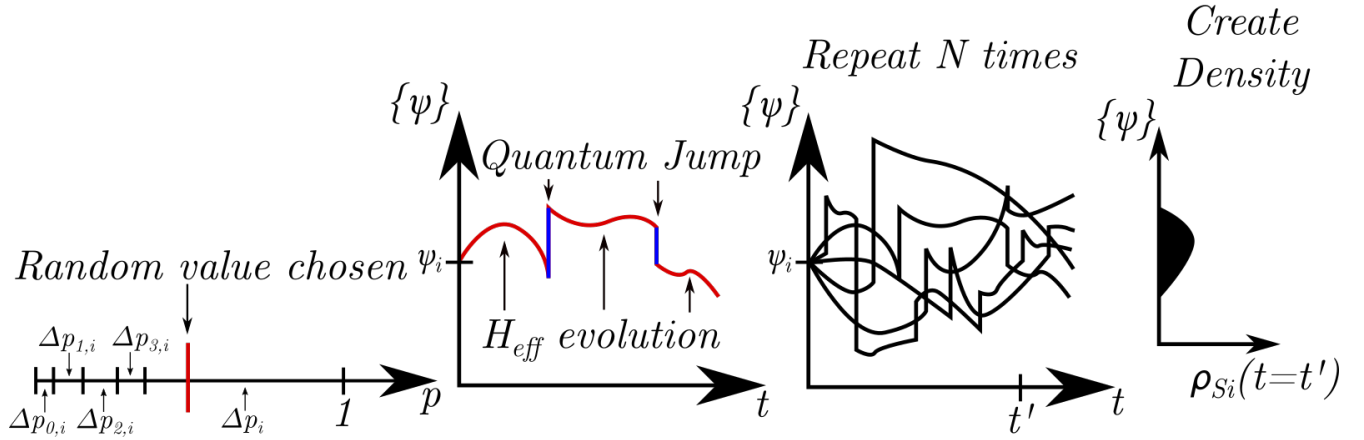
- Pick initial state $|\psi_i\rangle$
- Pick a random number $p \in [0, 1]$
- For a time interval Δt calculate Δp_i
 - If $\Delta p_i < p$ replace $|\psi_i(t)\rangle \rightarrow |\tilde{\psi}_i(t + \Delta t)\rangle$
 - If $\sum_k = 1^K \Delta p_{k,i} \leq p < \sum_k = 1^{K+1} \Delta p_{k,i}$ replace $|\psi_i(t)\rangle \rightarrow |\tilde{\phi}_i(t + \Delta t)\rangle$
- Repeat steps 2 and 3 for the desired length of times
- Repeat steps 1-4 N times to generate a distribution of outcomes (probability distribution)
- If necessary repeat steps 1-5 for all possible initial states $|\psi_i\rangle$
- Now we can at each timestep construct the density $\rho_S(t) = \frac{1}{N} \sum_i p_i |\psi_i\rangle \langle \psi_i|$

Figure 3 shows an illustration of this procedure.

3.1 The two level system

With all the tools we need, we are now able to describe the two level system, from the section on the interpretation of the Lindblad equation, with an effective Hamiltonian,

$$\begin{aligned} H_{eff} &= \hbar\omega \left(a_+ a_- + \frac{1}{2} \right) + \hbar\lambda (\sigma_+ + \sigma_-) - \frac{\hbar\gamma}{2} ((\bar{n} + 1) \sigma_-^\dagger \sigma_- + \bar{n} \sigma_+^\dagger \sigma_+) \\ &= \hbar\omega \left(a_+ a_- + \frac{1}{2} \right) + \hbar\lambda (\sigma_+ + \sigma_-) - \frac{\hbar\gamma}{2} ((\bar{n} + 1) \sigma_+ \sigma_- + \bar{n} \sigma_- \sigma_+). \end{aligned} \quad (56)$$



This system could be thought of as an atom in a box. The atom has two levels, which act as a quantum bit. We interact with the atom (and our quantum bit) by using a light field with strength λ . We now imagine that the walls of the box have a certain temperature and thus act as an environment which can interact with our atom through black-body radiation, thus making it an open quantum system (see Figure 2).

We will only consider one initial state $|\psi_2\rangle$. That means our entire population starts in the upper state. There are two operators $L_+ = \gamma\bar{n}\sigma_+$ and $L_- = \gamma(\bar{n} + 1)\sigma_-$ and thus we also have two jump states. Therefore, for each timestep we need to compute two jump probabilities, $\Delta p_{+,i}$ and $\Delta p_{-,i}$. The procedure of the project from here would be:

- Make a Monte Carlo simulation of the two-level system discussed, with and without the driving field (the $\hbar\lambda(\sigma_+ + \sigma_-)$). You can also try making the driving large $\lambda \gg \gamma$ or the other way around.
- Plot the density evolution of $|\psi_1\rangle$ and $|\psi_2\rangle$
- Interpret the results
- (Optional) try only having your light action turned on for $\lambda t = \pi/2$ starting at any point. This should transfer the entire population from $|\psi_2\rangle \rightarrow |\psi_1\rangle$ if there was no environment. But since the environment scrambles our state, we might see different results if we wait before we pulse our light interaction.

If you get completely stuck ChatGPT is more than capable of producing a code which does exactly this. However, try asking it to produce *pseudo code* for you instead. Then it produces an outline with functions and where to use them, but you have to fill out the rest.